Fangqiang Ding

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EDUCATION

Tongji University, Shanghai **September, 2017-June, 2021**

B.Eng in Mechanical Engineering ***GPA**: 4.77/5.00 or 92.7/100 (1/129)*

— **Scholarship**: Double National Scholarship (1%) and double Excellent Student Scholarship

— **Supervisor**: Prof. Changhong Fu (Director of **Vision4Robotics Group**)

Tsinghua University, Beijing **August, 2020-September, 2020**

Visiting Student in Department of Automation

— **Supervisor**: Prof. Geng Lu (Director of UAV Lab)

RESEARCH INTERESTS

Present: Unmanned Aerial Vehicle, Computer Vision, Object Tracking

Future: Robotics, Pose Estimation, 3D Computer Vision

RESEARCH EXPERIENCES

Improving Object Tracking Accuracy in UAV Scenarios by Regularization Modification

June, 2019-November, 2019

- Exploited the automatic adaptation of spatio-temporal regularization for correlation filters optimization. Local and global properties of detection-used response matrix are adopted to elevate the robustness of trackers. (*accepted by CVPR 2020*)
- Proposed multi-view selection mechanism to realize memory augmentation for correlation filters to alleviate exponential decay of information. Perception hash algorithm is adopted to calculate the similarity of patches and decide whether to update the memory. (*accepted by IROS 2020*)

Investigating Real-Time and Robust Methods for Aerial Tracking on a Single CPU

November, 2019-Present

- Presented distractor repressed dynamic regression tracker. The local extreme points in response map are used to localize the distractors and the regression label of objective function is dynamically adjusted by pixel-level penalization of distractors. (*accepted by IROS 2020*)
- Proposed joint scale and aspect ratio optimization by training separate filer in size domain. Inspired by two-stage object detection methods, a novel object proposal based re-detection strategy is

presented to cope with long-term tracking tasks. (*accepted by IROS 2020*)

- Investigated appearance variation information to adaptively generate a spatial reliability map. A response pool is proposed as a standard of comparison to measure the stability of response map and influence its spatial reliability. (*submitted to IEEE Trans. Multimedia*)

Exploring Deep Learning Methods for Visual Object Tracking **May, 2019-Present**

- Investigated integration of two complementary types of methods, *i.e.,* end-to-end trackers and correlation filter-based approaches to obtain CPU-friendly and high-performance trackers.
- Proposed to train domain translation methods based on Generative Adversarial Networks (GANs) to generate multi-modal (RGB-T or RGB-D) tracking datasets using original visible spectrum data.
- Applied deep learning-based low-light image enhancement methods to visual object tracking tasks for performance improvement under dark scenes. A novel benchmark used for evaluating trackers in dark scenes is currently made.
- Presented guider-anchors method to solve unbalance positive and negative samples problem. Adopting this strategy, Siamese-based trackers can achieve fast speed (over 300fps) without casting aside intrinsic excellent performance.

Vision-Based UAV Localization in Air-Ground Robots Coordination **May, 2019-Present**

- Deigned a benchmark system that enables the evaluation of visual object tracking methods on UAV localization under air-ground robot coordination scenarios. Specifically, ground-truth positions were obtained from motion capture system and the estimated results were output from ROS.
- Validated the effectiveness of various object tracking in accurate localization of drone. Visual trackers are assigned to predict markers' positions in image domain while classical framework of UAV pose estimation is adopted to estimate the trajectory of UAV in world frame.

PUBLICATION

Conference

- Yiming Li, Changhong Fu, **Fangqiang Ding**, Ziyuan Huang and Geng Lu. *AutoTrack: Towards High-Performance Visual Tracking for UAV with Automatic Spatio-Temporal Regularization.*, in IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR'20**). Published.
- **Fangqiang Ding**, Changhong Fu, Yiming Li, Jin Jin and Chen Feng. *Automatic Failure Recovery and Re-Initialization for Online UAV Tracking with Joint Scale and Aspect Ratio Optimization*, in IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS'20**). Accepted.
- Changhong Fu, **Fangqiang Ding**, Yiming Li, Jin Jin and Chen Feng. *DR^{2}Track: Towards Real-Time Visual Tracking for UAV via Distractor Repressed Dynamic Regression*, in IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS'20**). Accepted.
- Yiming Li, Changhong Fu, **Fangqiang Ding**, Ziyuan Huang and Jia Pan. *Augmented Memory for*

Correlation Filters in Real-time UAV Tracking, in IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS'20**). Accepted.

Journal

- Changhong Fu, **Fangqiang Ding**, Yiming Li, Jin Jin and Chen Feng. *Learning Dynamic Regression with Automatic Distractor Repression for Real-Time UAV Tracking*, submitted to Engineering Applications of Artificial Intelligence. Under review.
- Changhong Fu, Jin Jin, **Fangqiang Ding**, Yiming Li and Geng Lu. *Spatial Reliability Enhanced Correlation Filter: An Efficient Approach for Real-Time UAV Tracking*, submitted to IEEE Transactions on Multimedia. Under review

Magazine

- Changhong Fu, Bowen Li, **Fangqiang Ding**, Fulin Li and Geng Lu. *Correlation Filter for UAV-Based Aerial Tracking: A Review and Experimental Evaluation*, submitted to IEEE Geoscience and Remote Sensing Magazine. Under review.

Selected Projects

Zeal Eco-Power Racing Vehicle Team **October, 2018-Present**

- Participate in designing vehicle velocity measurement and oil cut-off system.
- Developed mobile APP for the visualization of vehicle state parameters and driving control.
- Equipped racing vehicles with perception ability using computer vision algorithms.

Developing Intelligent Following Umbrella **January, 2019-May, 2019**

- Proposed to calculate pedestrian position by UWB multi-module communication.
- Exploited controlling motors directly to simulate the remote controller.
- Incorporated our designed novel umbrella for implementation.

ACADEMIC SERVICE

IROS 2020	Reviewed three submissions
IEEE JSTSRS	Reviewed one submission
IEEE TCYB	Reviewed one submission
IEEE TPAMI	Reviewed one submission

AWARDS

- First Prize of Tongji Undergraduate Mathematics Competition (top 10) **June, 2018**
- First Prize of Shanghai College Student Innovative Design Competition (top6) **May, 2018**
- Second Prize of Shanghai College Student Physics Competition **December, 2018**
- Second Prize of Tongji Undergraduate Physics Competition **May, 2018**
- Third Prize of Chinese Undergraduate Mathematics Competition **October, 2018**

- Third Prize of Chinese Undergraduate Mathematics Modelling Competition **November, 2018**
- Third Prize in Shanghai Mechanics Competition **May, 2019**

KEY SKILLS

Programming Language	Matlab, Python, C++, Java, HTML
Developing&Simulation Tools	Keil, Linux, ROS, Android Studio, Multisim LaTex, Adobe Premiere Pro, AutoCAD, Solidworks, Autodesk Inventor